Southeast University

Department of Computer Science and Engineering

Course Name: Database Design Lab

Course Code: CSE384

Lab No: 03

Lab Title: Table Creation and Data Manipulation

Date of Submission: 24/11/2025

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Course: CSE384

Section: 07

Objective: The objective of this lab session was to practice basic SQL operations related to database creation and table manipulation. The tasks included creating a database, creating tables with constraints, inserting data, updating records, deleting rows, and modifying table structures using ALTER TABLE commands.

Problems and Solutions:

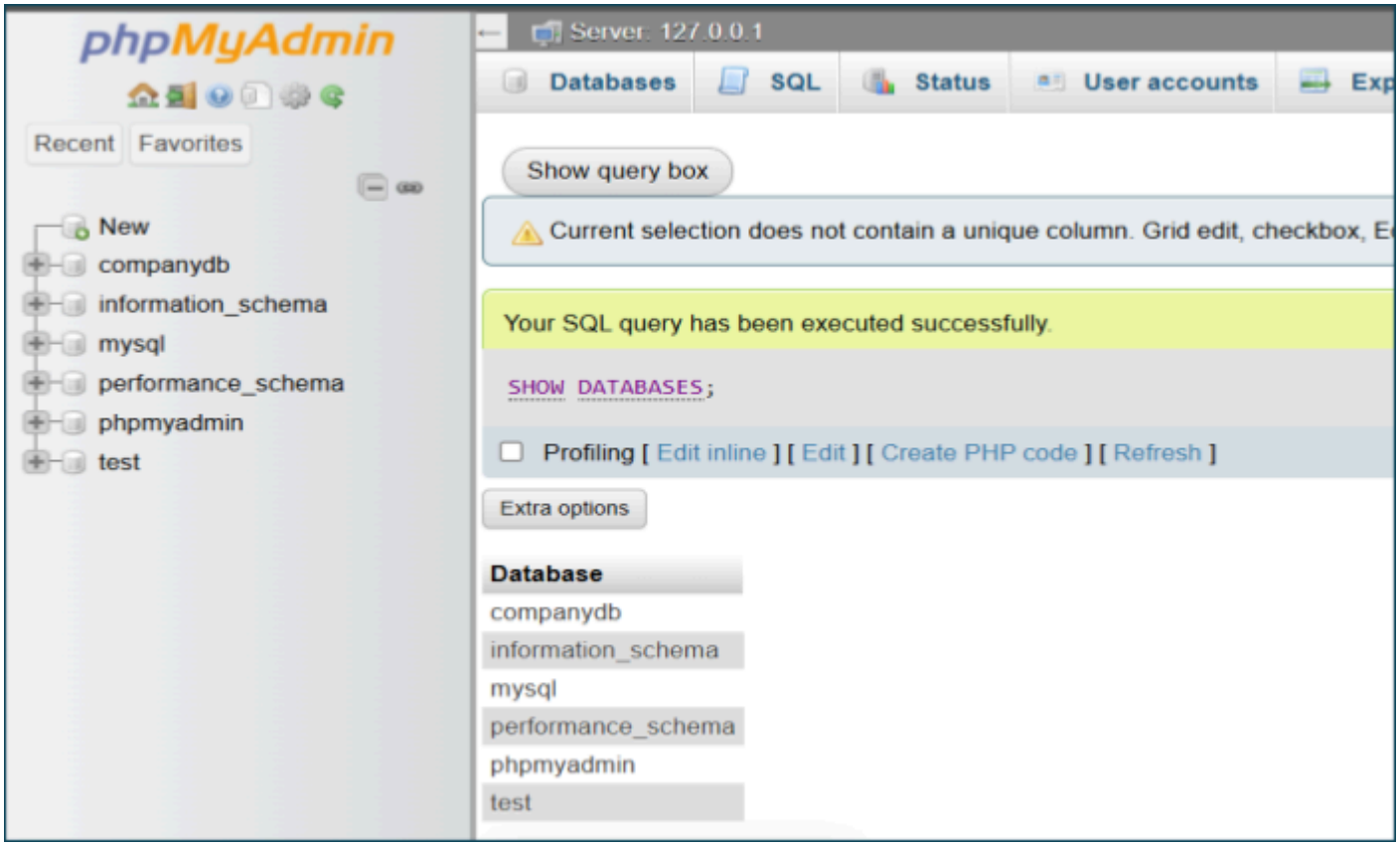
Problem 1:

Title: Create a database named CompanyDB and verify it

SQL Query: `CREATE DATABASE CompanyDB;`

`SHOW DATABASES;`

Output:



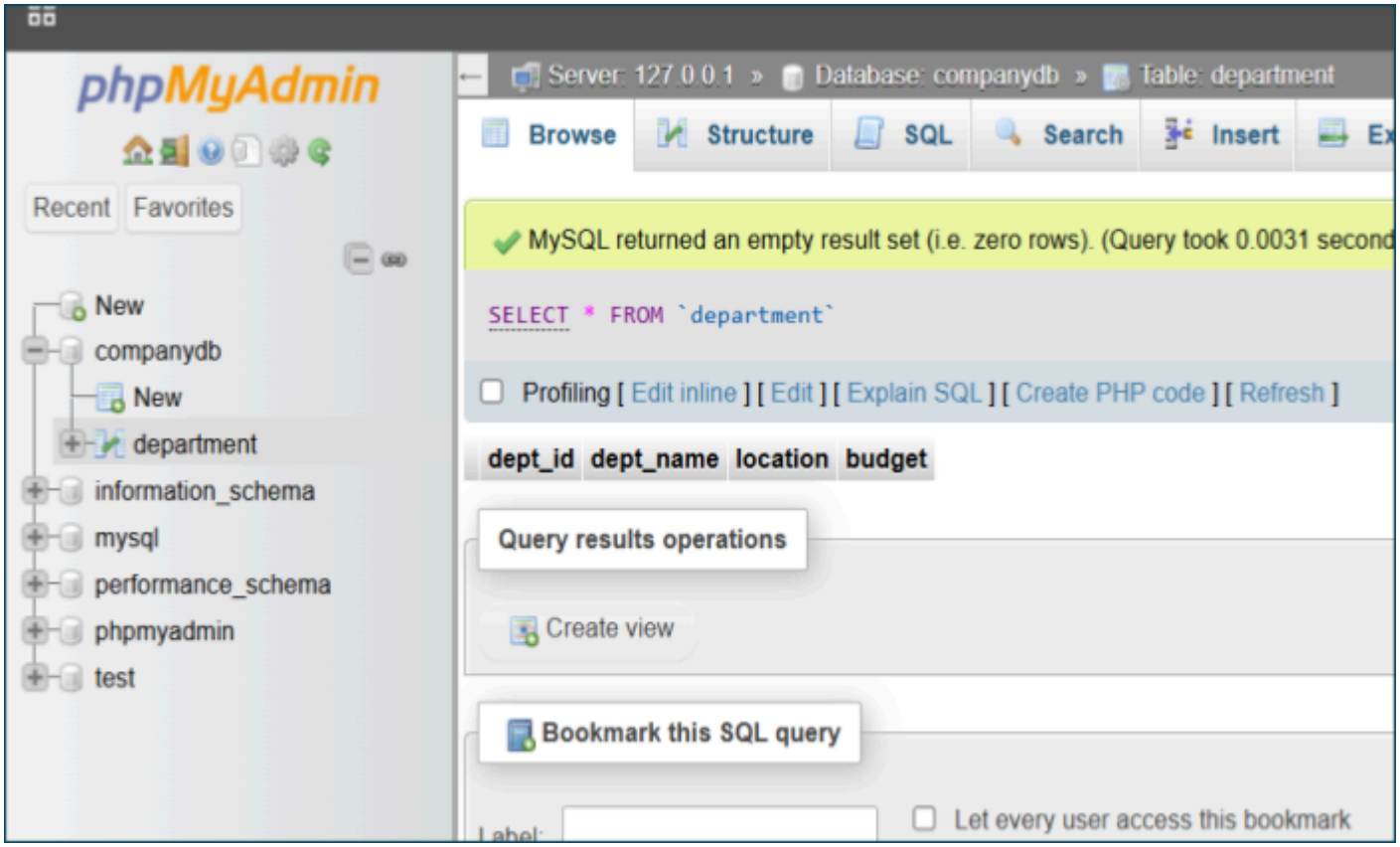
Problem 2:

Title: Create the Department table with dept_id as Primary Key and budget with a CHECK constraint (min 100000).

SQL Query:

```
CREATE TABLE Department ( dept_id INT PRIMARY KEY, dept_name VARCHAR(50) NOT NULL, locati
on VARCHAR(50), budget INT CHECK (budget >= 100000) );
```

Output:



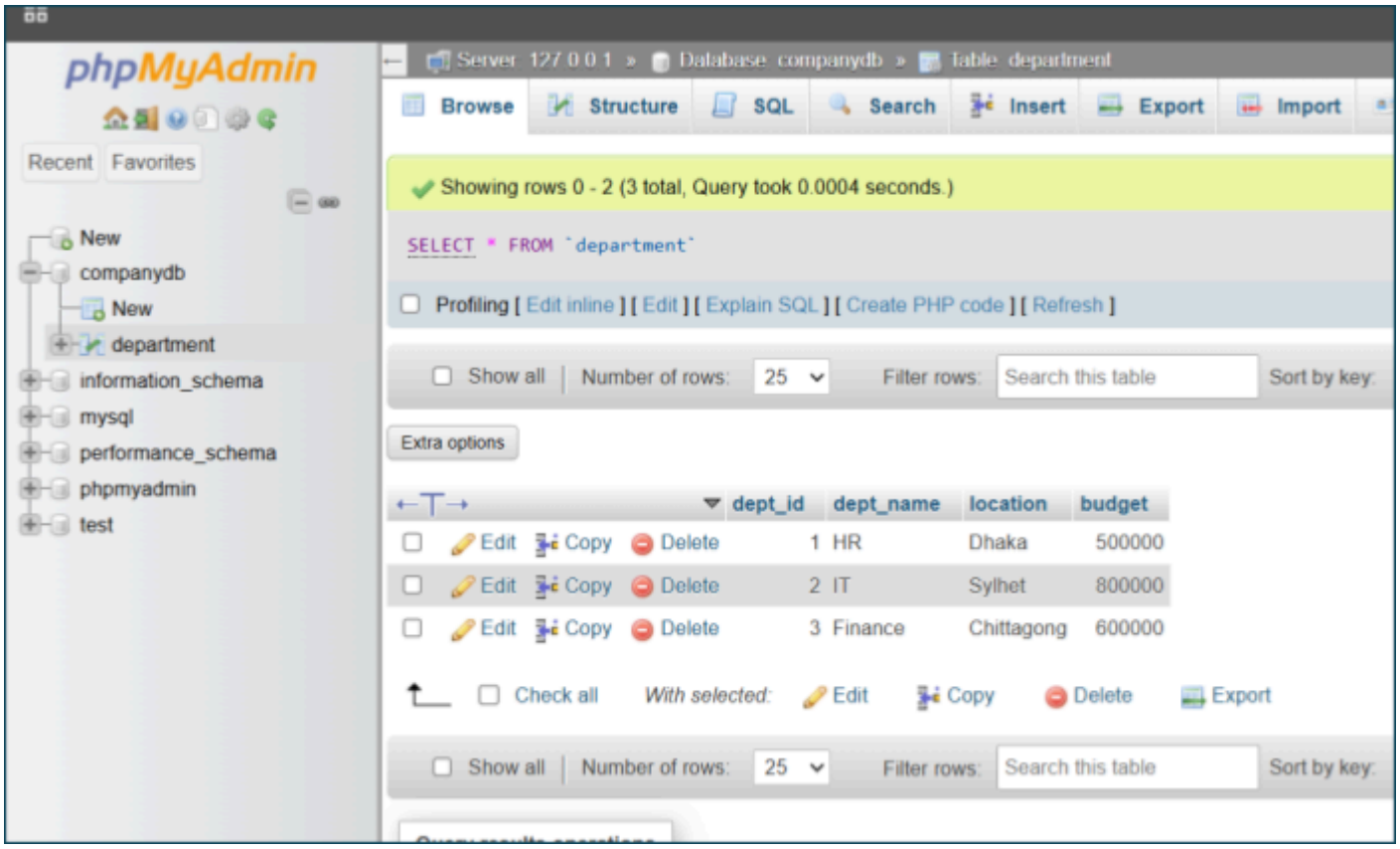
Problem 3:

Title: Insert sample data into the Department table.

SQL Query:

```
INSERT INTO Department (dept_id, dept_name, location, budget) VALUES (1, 'HR', 'Dhaka', 500000), (2, 'IT', 'Sylhet', 800000), (3, 'Finance', 'Chittagong', 600000);
```

Output:

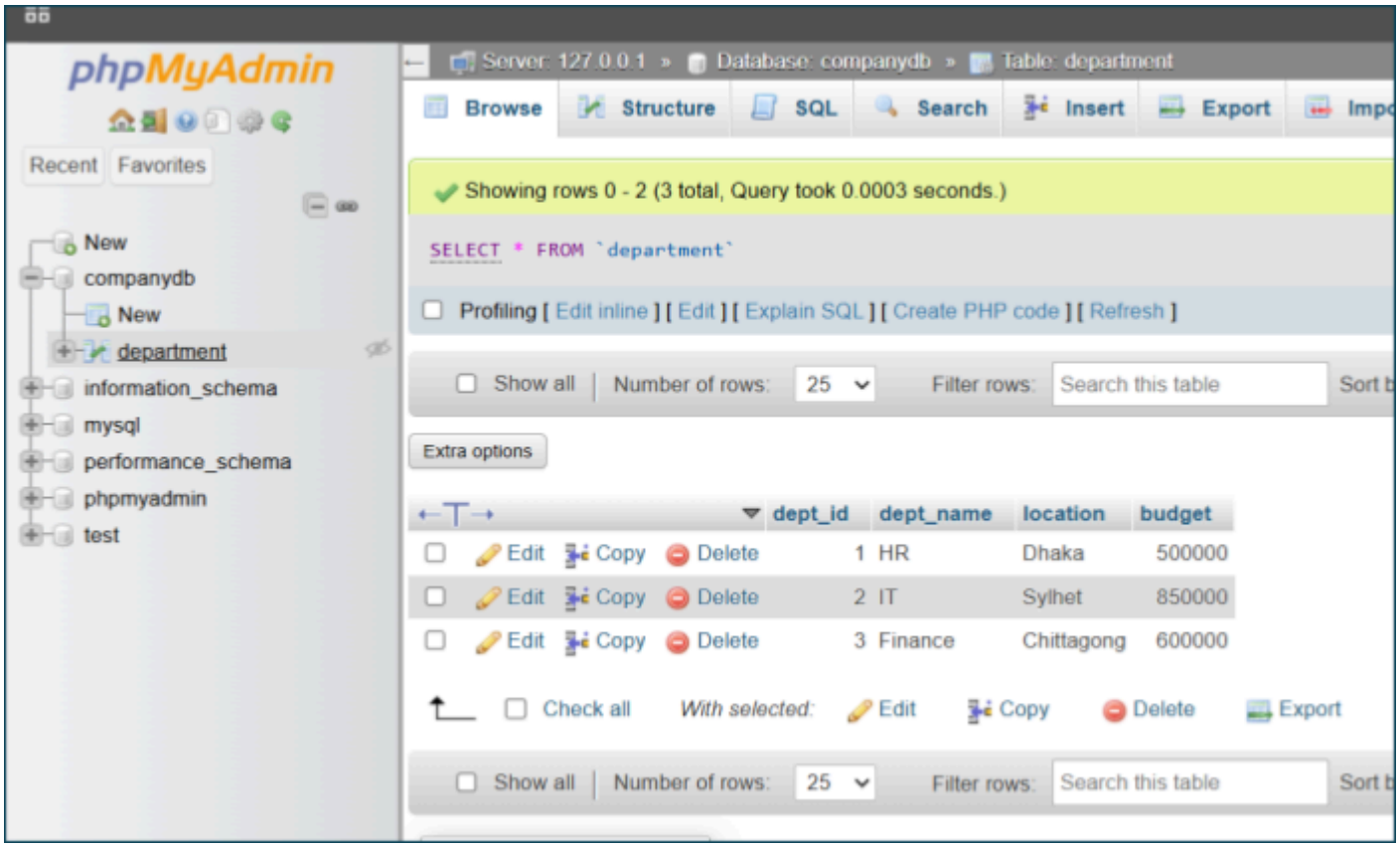


Problem 4:

Title: Update the budget of IT to 850000.

SQL Query: `UPDATE Department SET budget = 850000 WHERE dept_id = 2;`

Output:

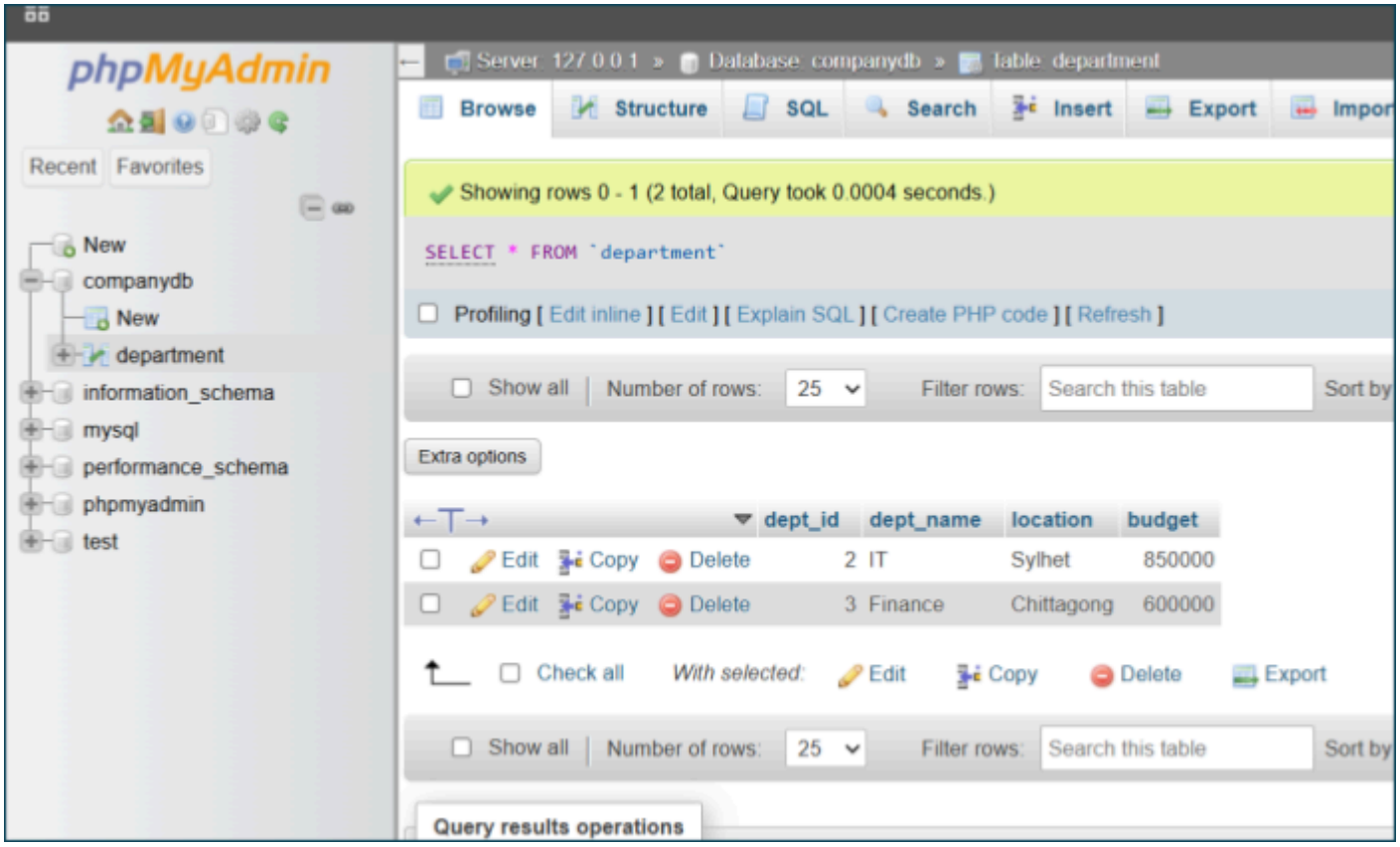


Problem 5:

Title: Delete a department with a budget of less than 600000.

SQL Query: `DELETE FROM Department WHERE budget < 600000;`

Output:

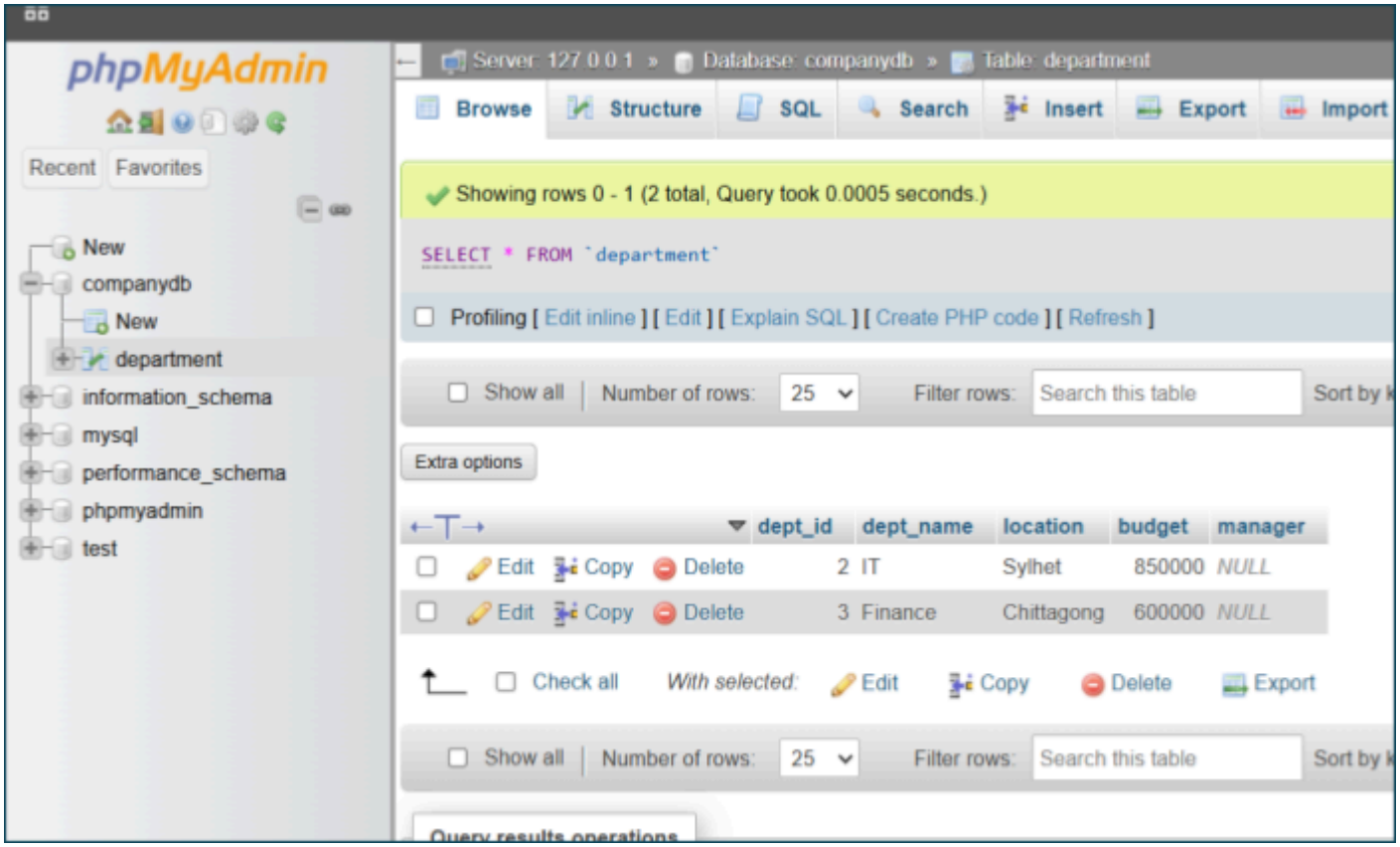


Problem 6:

Title: Add a 'manager' column to the Department table.

SQL Query: `ALTER TABLE Department ADD (manager VARCHAR(50));`

Output:



Problem 7:

Title: Modify the location column VARCHAR(100)

SQL Query: `ALTER TABLE Department MODIFY location VARCHAR(100);`

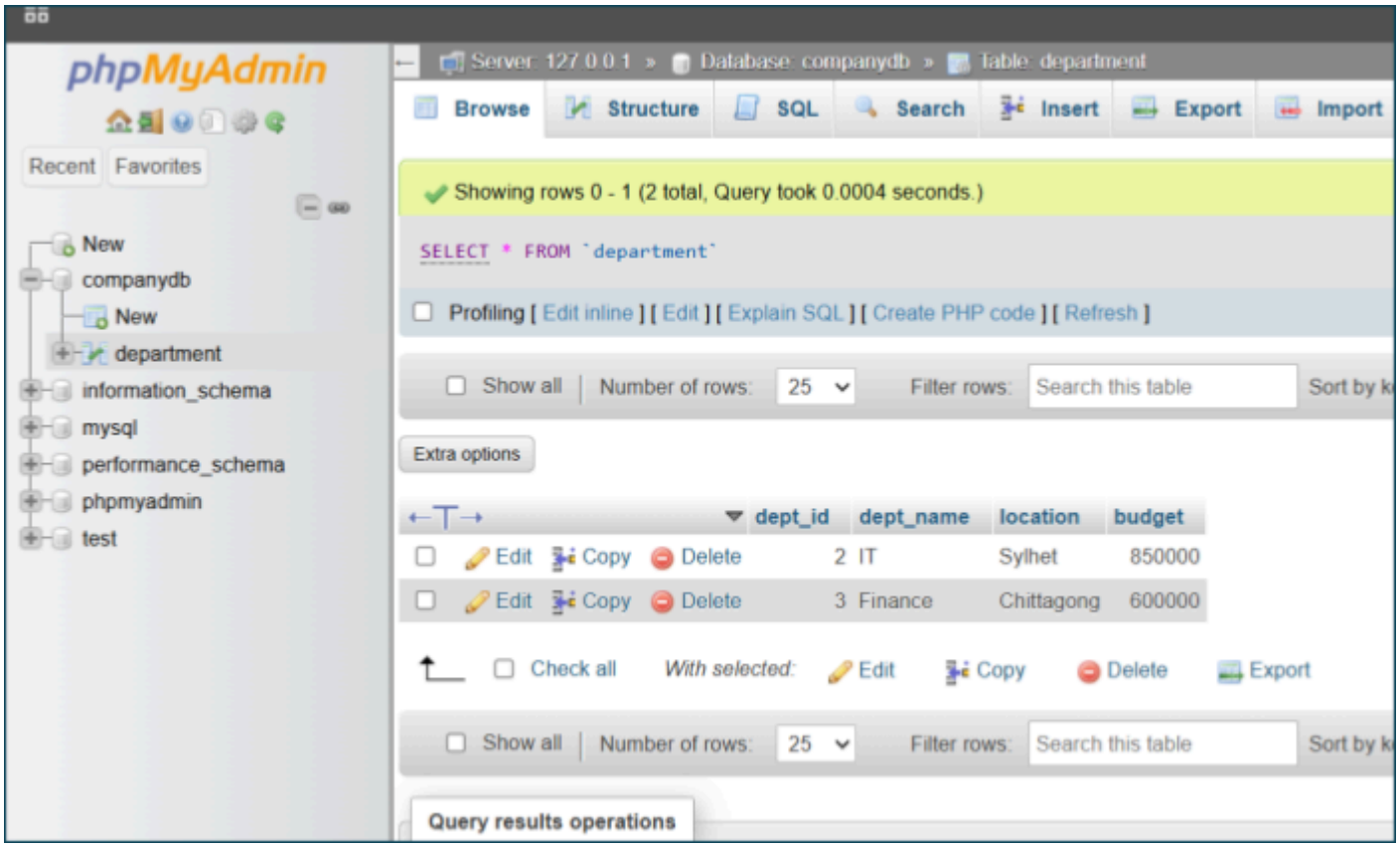
Output:

Problem 8:

Title: Drop the manager column.

SQL Query: `ALTER TABLE Department DROP COLUMN manager;`

Output:



Problem 9:

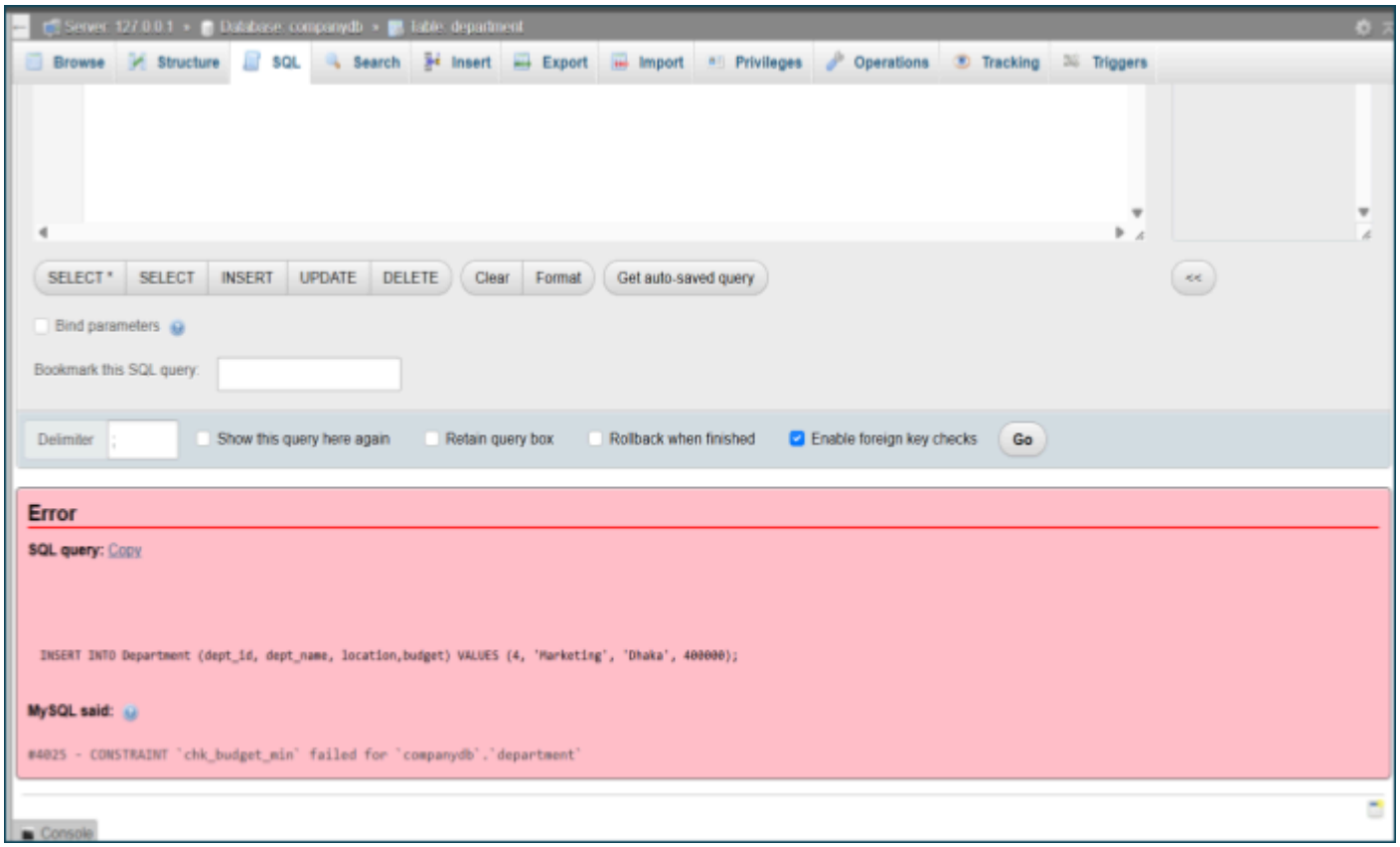
Title: Add a constraint on the budget column.

SQL Query:

```
ALTER TABLE Department ADD CONSTRAINT chk_budget_min CHECK (budget >= 500000);
```

```
INSERT INTO Department (dept_id, dept_name, location,budget) VALUES (4, 'Marketing', 'Dhaka', 400000);
```

Output:



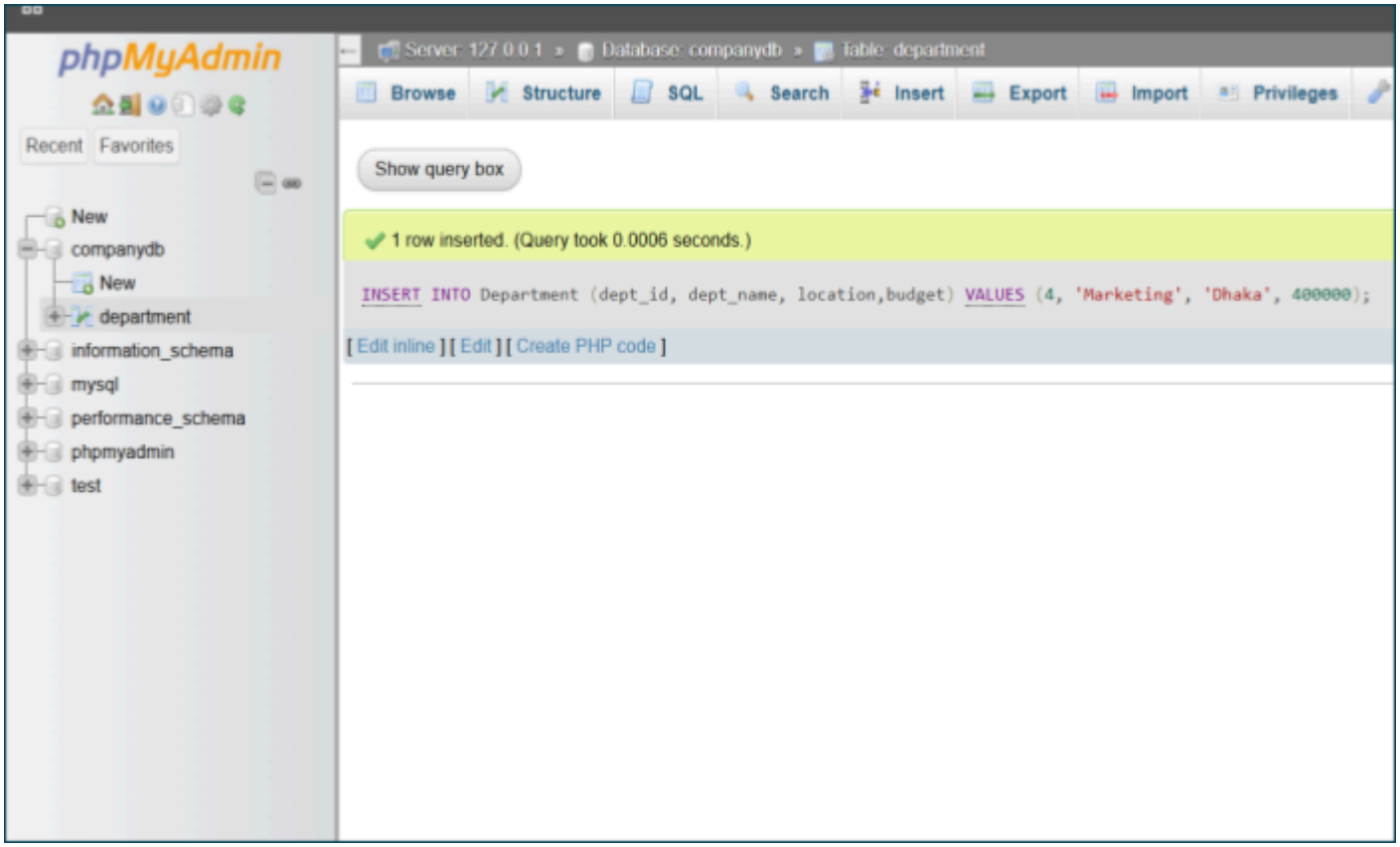
Problem 10:

Title: Drop the budget constraint.

SQL Query: `ALTER TABLE Department DROP CONSTRAINT chk_budget_min;`

```
INSERT INTO Department (dept_id, dept_name, location,budget) VALUES (4, 'Marketing', 'Dhaka', 400000);
```

Output:

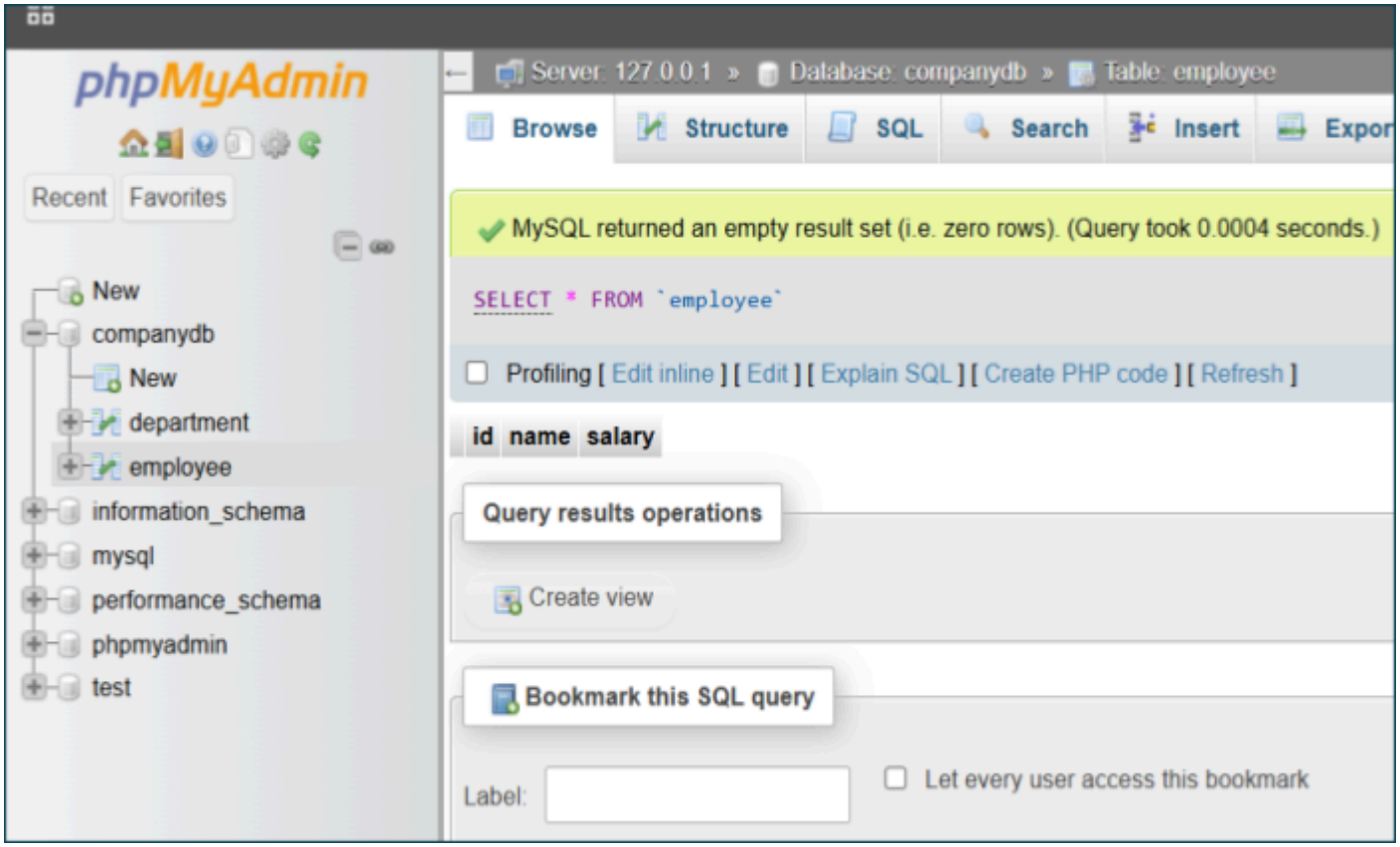


Problem 11:

Title: Create the Employee table with emp_id.

SQL Query: `create table employee(id INT PRIMARY KEY, name VARCHAR(50), salary int);`

Output:



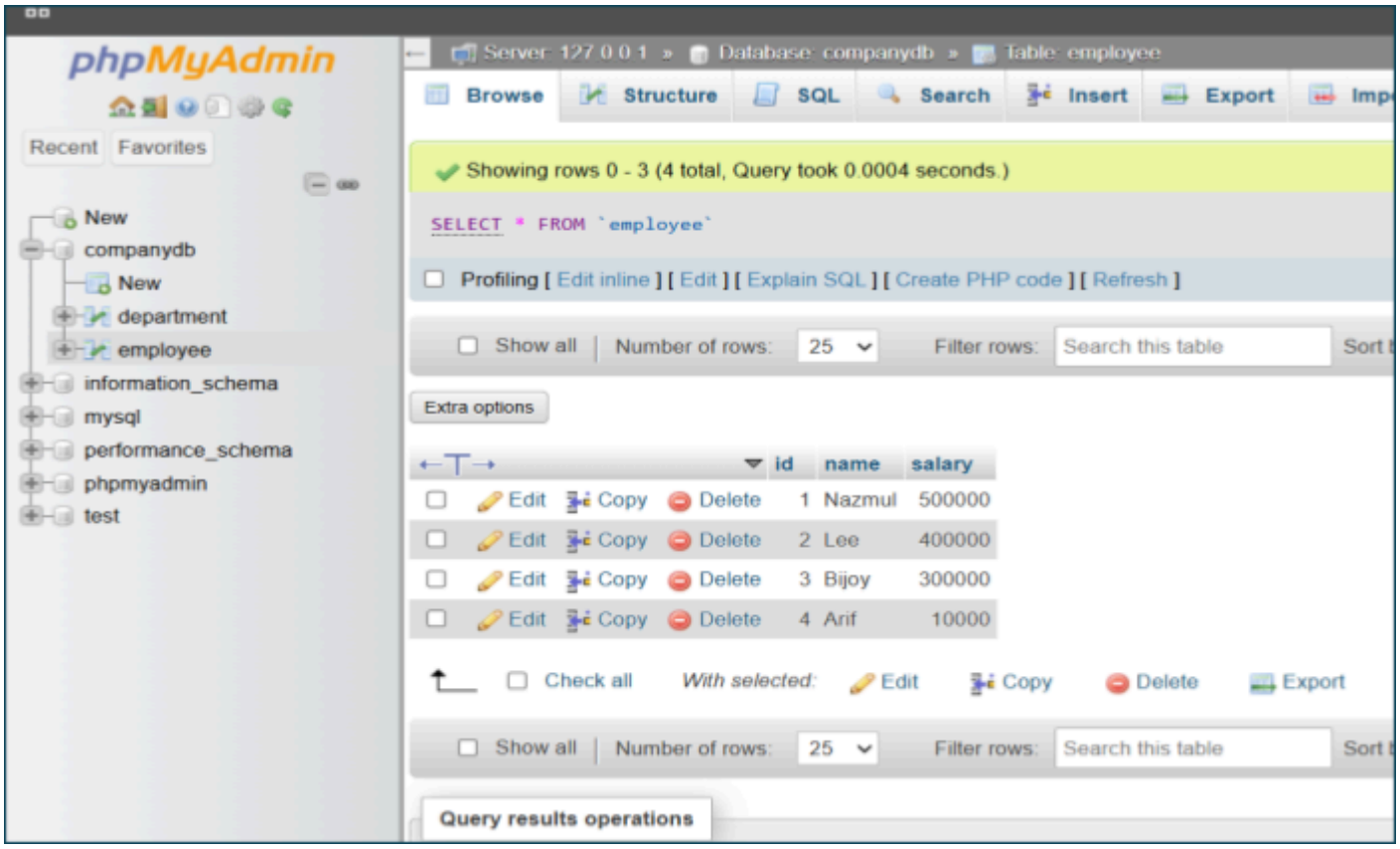
Problem 12:

Title: Insert 4 rows into Employee.

SQL Query:

```
INSERT INTO employee (id, name, salary) VALUES (1, 'Nazmul', 500000), (2, 'Lee', 400000),
 (3, 'Bijoy', 300000), (4, 'Arif', 10000);
```

Output:

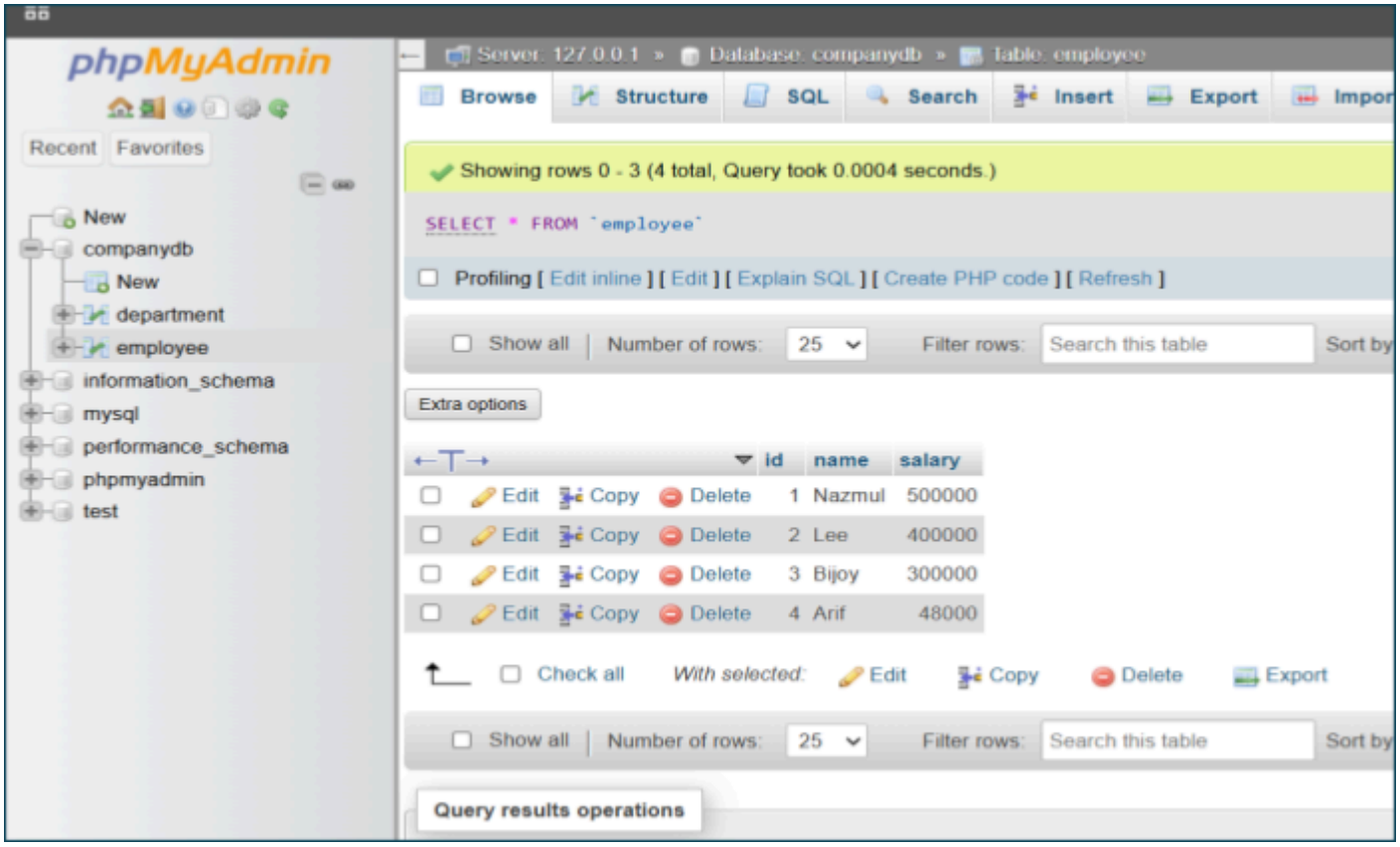


Problem 13:

Title: Update the salary of Arif to 48000.

SQL Query: `UPDATE employee SET salary = 48000 WHERE name = 'Arif';`

Output:

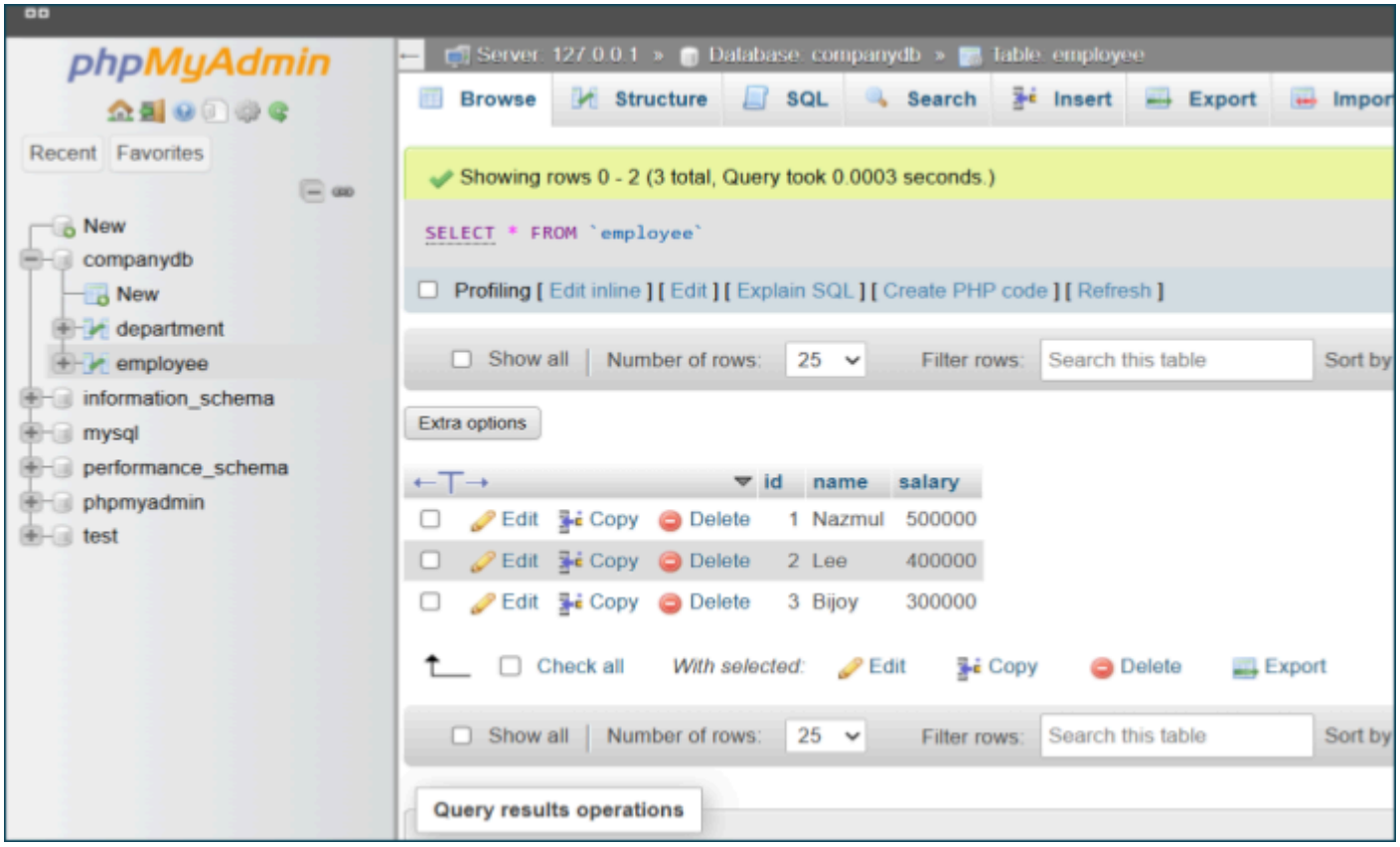


Problem 14:

Title: Delete an employee with a salary of less than 50000.

SQL Query: `DELETE FROM employee WHERE salary < 50000;`

Output:

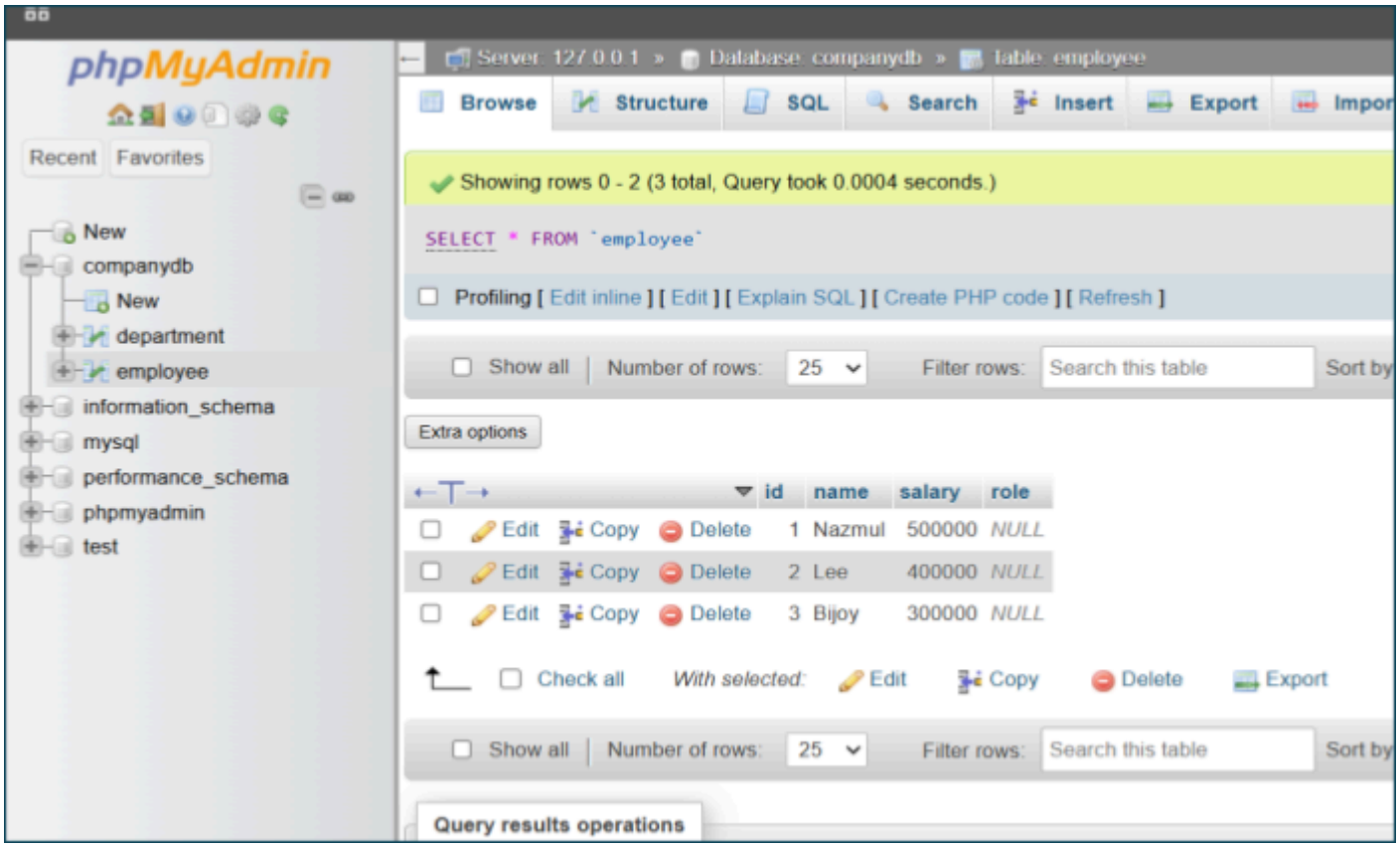


Problem 15:

Title: Add a 'role' column to Employee (e.g., 'Junior' or 'Senior').

SQL Query: `ALTER TABLE employee ADD (role VARCHAR(50));`

Output:

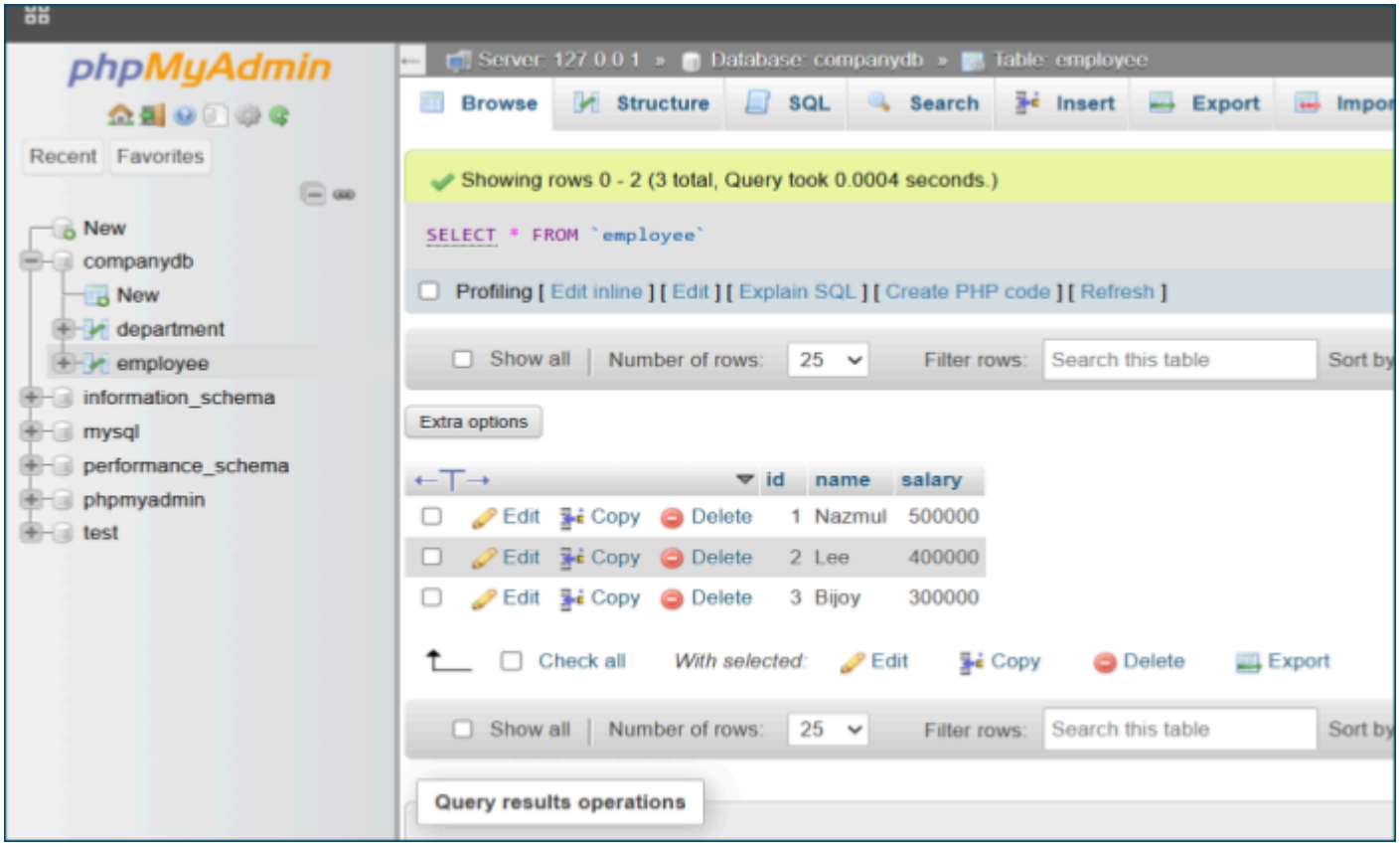


Problem 16:

Title: Drop the role column.

SQL Query: `ALTER TABLE employee DROP COLUMN role;`

Output:



Result: Throughout this lab, the fundamentals of SQL table creation and data manipulation were successfully practiced. Operations such as creating databases, inserting and updating records, applying constraints, modifying columns, and deleting data were performed. The expected outputs matched the executed SQL queries, confirming correct understanding of SQL commands.

Discussion: During the lab, various SQL concepts were reinforced, including primary keys, CHECK constraints, and ALTER TABLE operations. One key observation was how constraints prevent invalid data entry, ensuring data integrity. Errors could occur if syntax was incorrect or if inserted values violated constraints. The session improved understanding of how structural and data related changes affect the table and database.

Conclusion: This lab enhanced practical skills in SQL by providing hands on experience in database and table creation, applying constraints, modifying table structures, and managing data. The tasks strengthened foundational knowledge required for designing and maintaining relational databases in future coursework and industry level applications.